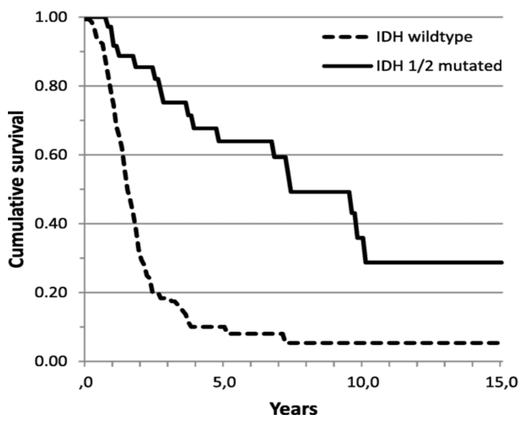
A Benchmark of (MRI-) Foundation Models to Predict IDH Mutational Status in Glioma

Nathan Hollet*[1,2], Elise Robinson*[1,3], Efthymios Georgiou[1,2], Uri Nahum[4,5], Ekin Ermis[1], Sarah Brueningk[1,2]

[1]Department of Radiation Oncology, Inselspital, Bern University Hospital and University of Bern, Switzerland, [2]Department of Digital Medicine, University of Bern, Switzerland
[3]School of Life Sciences, FHNW University of Applied Sciences and Arts, Muttenz, Switzerland, [4]School of Business, FHNW University of Applied Sciences and Arts, Olten, Switzerland
[5]University Children's Hospital Basel UKBB, Basel, Switzerland
*equal contribution

MOTIVATION & CLINICAL QUESTION



In glioblastoma, IDH-wildtype patients have a median survival of 1.6 years versus 7.5 years for IDH-mutant, a **~4.7× difference invisible on MRI**, yet crucial for prognosis, and treatment selection [1].

\+ Part of clinical workflow
\+ Non-invasive
\- Low spatial resolution
\- Requires processing

MRI-based analysis



Baseline: Radiomics-based Approaches

New Methods: NEW Image Foundation Models

"Can large scale-pretrained image foundation models outperform established radiomics baselines for non-invasive IDH prediction in glioma? And how robustly does that performance transfer under cross-cohort distribution shift?"

DATA & MODELS

pre Pre-treatment datasets
post External post-treatment dataset

Cohort	N	WHO Grades (2/3/4)	%IDH-mut
UCSF-PDGM[pre]	494	56 / 43 / 395	21%
UPENN-GBM[pre]	515	0 / 0 / 515	3.1%
EGD[pre]	467	127 / 29 / 259	33%
UTSW-Glioma[pre]	619	11 / 97 / 416	28%
UCSD-PTGBM[post]	123	0 / 0 / 123	9.8%

***Table 1**: Cohort summary. All cohorts use T2-FLAIR+T1 post-contrast.*

Model	Param.	Training Data	Objective	Input vec.
BrainIAC [6]	88.3M	32k brain MRI	SimCLR	1536
MRI-CORE [3]	85.8M	110k MRI	DINOv2	1536
BiomedCLIP [4]	86.2M	15M image-text pairs	CLIP	3072
BrainDINO [5]	85.6M	6.6M brain MRI slices	DINOV3	4608
TabPFN [2]	~28M	Synth. tabular tasks	PFN	214

***Table 2**: The five foundation models benchmarked in this study. All four image encoders are ViT-B/16 variants (BrainIAC operates in 3D; MRI-CORE is the SAM image encoder), while TabPFN is a tabular transformer. Input vec. denotes the per-patient FLAIR + T1 vector dimension fed to the linear probe.*

METHODS

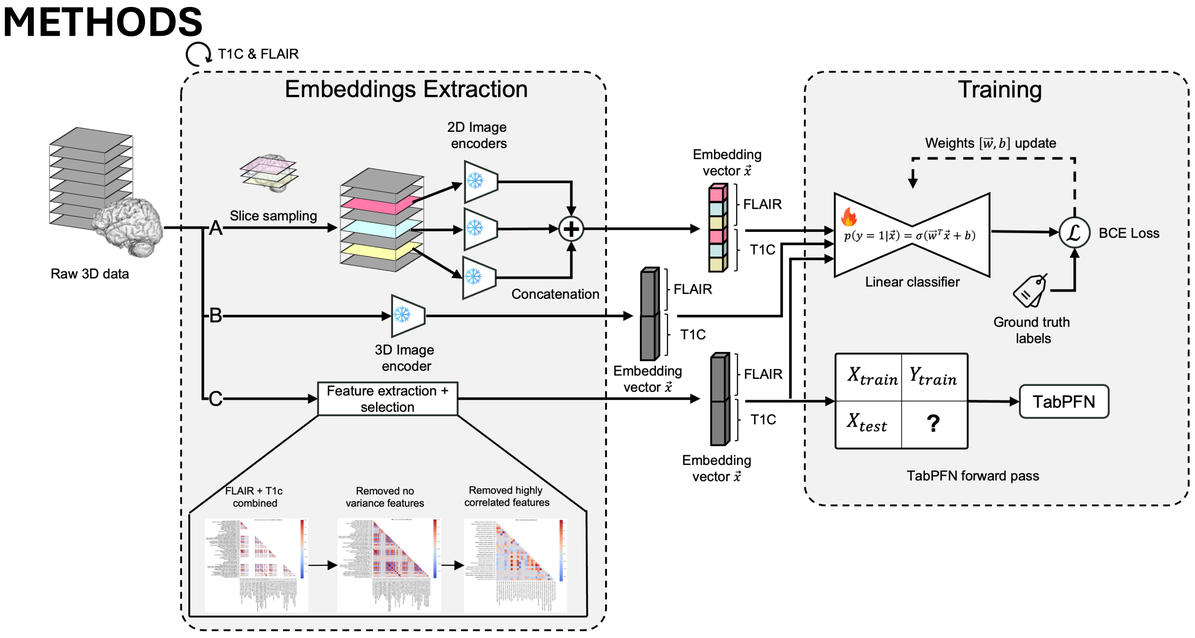


***Figure 1**: Schematic overview of the processing and training pipeline. Each scan (FLAIR and post-contrast T1) feeds three parallel frozen pipelines and a single trainable classifier on top. **(A)** 2D image foundation models (MRI-CORE, BiomedCLIP, BrainDINO) take three axial slices at the 25th, 50th, and 75th percentile of the tumor mask and concatenate their CLS tokens. **(B)** The 3D image foundation model (BrainIAC) consumes the full skull-stripped volume. **(C)** PyRadiomics features are extracted from the segmented tumor with per-fold feature selection. All vectors feed a trainable linear probe; the radiomics vector additionally feeds TabPFN in-context, with no gradient updates.*

RESULTS

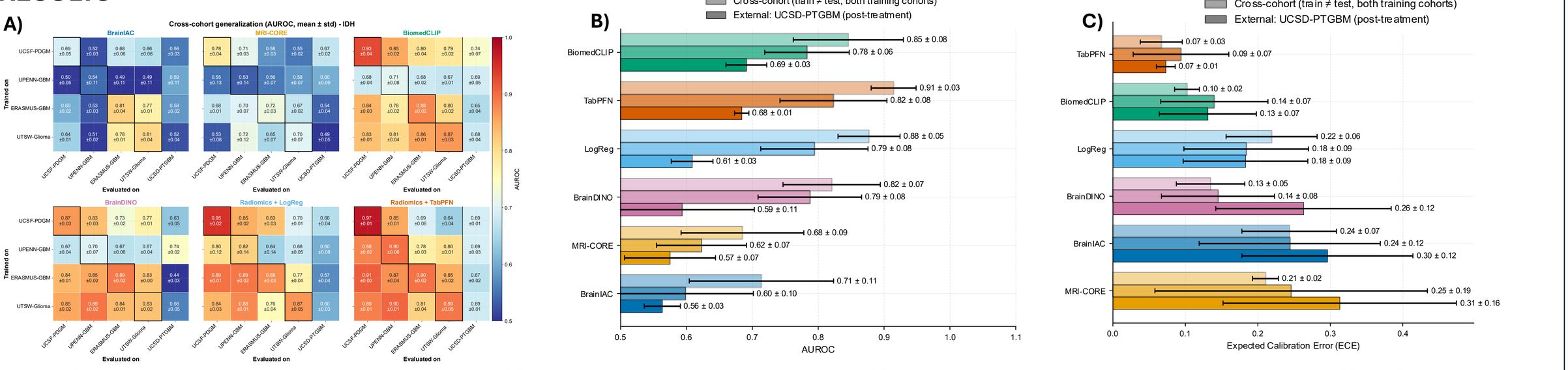


***Figure 2**: Cross-cohort generalization summary: **(A)** per-model AUROC heatmap across all train × test pairs, **(B)** AUROC ranking under in-distribution, cross-cohort, and external (UCSD-PTGBM) settings; higher is better, **(C)** ECE (Expected Calibration Error): how closely a model's predicted probabilities match the observed outcome rate; lower is better.*

CONCLUSION

TabPFN [2] matches or exceeds every image foundation model on AUROC in- and cross-cohort and remains the **best-calibrated baseline in every setting**, including the post-treatment external. **BiomedCLIP [4] only takes the lead on AUROC under heavy shift**, never on calibration, while the older brain-specific encoders (BrainIAC [6], MRI-CORE [3]) consistently underperform. Together, these findings suggest that **representation choice for non-invasive IDH prediction should be guided by clinical context**, and that **combining radiomics and image-foundation-model features is a promising direction**.



Nathan Hollet
nathan.hollet@unibe.ch

Center for AI in Radiation Oncology
Department of Radiation Oncology,
Inselspital and University of Bern

Swiss National Science Foundation

References:

1) Synhaeve et al. Clinical evaluation of a dedicated next generation sequencing panel for routine glioma diagnostics.
2) Hollmann, N. *et al.* TabPFN: a transformer that solves small tabular classification problems in a secon
3) Dong, H., Chen, Y., Gu, H., Konz, N., Chen, Y., Li, Q., Mazurowski, M.A.: MRI-CORE: A Foundation Model for Magnetic Resonance Imaging
4) Zhang, S., et al.: BiomedCLIP: a multimodal biomedical foundation model pretrained from millions of image-text pairs
5) Wu, Y., Wang, S., Li, Y., Safari, M., Hu, M., Chang, C.W., Veeraraghavan, H., Yang, X.: Braindino: A brain mri foundation model for generalizable clinical representation learning
6) Tak, D., et al.: A generalizable foundation model for analysis of human brain MRI